V - Vector space

↳ how do we create $W < V$

↑ Subspace

[you have to be creative] ← practice

Create W:

↳ pick out of any finite # of elements

$$\{V_1, V_2 \dots V_n\}$$

-> Why?

→ Suppose we have a homgenious system of L.E.

$$A \cdot \bar{X} = 0$$

$\dim A = m \times n$

$\dim(\bar{X}) = n \times 1$

$$\bar{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Clearly any solution $\in \mathbb{R}^n$

Theorem Solution space of hom S.L.E is a subspace of $\mathbb{R}^n$

↓ Proof

$A \cdot \bar{X} = \bar{0}$

$A \cdot \bar{Y} = \bar{0}$

$\bar{X}$ and $\bar{Y}$ are solutions

$W_1 \; W_2 \in W$

① $W_1 + W_2 \in K$

② $K \cdot W_1 \in W$

↳ – $\bar{X} + \bar{y}$ – (?) is it a solution?

↳ $A \cdot [\bar{X} + \bar{Y}] = A \cdot \bar{X} + A \cdot \bar{Y} = \bar{0} + \bar{0} = 0$

$\therefore \bar{X} + \bar{Y}$ is a solution

– multiplication by K

$A[K \cdot \bar{X}] = K[\quad]$

matrix multiply ↑ $\bar{0}$

=> proof is complete

② $A \cdot \bar{x} = \begin{bmatrix} \vdots \\ \vdots \end{bmatrix}$

find all solutions of above $\Rightarrow$ is it a subspace of $\mathbb{R}^n$

↳ any subspace must have 0 element $\rightarrow \bar{x} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$

$$A \cdot \bar{x} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \neq \begin{bmatrix} \vdots \\ \vdots \end{bmatrix}$$

$\Rightarrow$ it's only a subspace if S.L.E is homogenous

$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 3 & -6 & 9 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

So we know

W is Subspace

Solve this ↗

$x - 2y + 3z = 0$ ↩ $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Plane in $\mathbb{R}^3$, $P(0,0,0)$

$n = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

Liniar independence

V is a vector space

always given -> $\{\vec{v}_1, \vec{v}_2, \dots \vec{v}_n\} \in V$

are they liniar independent?

S.L.E

if $\boxed{c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n = \vec{0}}$

it has multiple solutions if $c_1 = c_2 = c_3 = \dots = c_n = 0$

Ex $V = \mathbb{R}^2$

↳ $V_1 \dots V_n$ → obviously $n \geq 2$

$n = 2$

$$V_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad V_2 = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$$

↳ $c_1\vec{V}_1 + c_2\vec{V}_2 = \vec{0}$

unknown from linear combination

↳ $\begin{bmatrix} \vec{V}_1 & \vec{V}_2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

↳ $\begin{bmatrix} 1 & 6 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

↳ solve

tricky part

1 - solution → YAY

∞ - solution → NAY

$V_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $V_2 = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$ $V_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} 1 & 6 & 1 \\ 2 & 7 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

↳ use augmented matrix approach

↳ $\begin{bmatrix} 1 & 6 & 1 & 0 \\ 2 & 7 & 0 & 0 \end{bmatrix}$ REF →

↓

blah

blah

blah